

Leakage Safe Similarity Neighbour Side Information for Graph Based Drug Target Interaction Prediction

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Abstract—Predicting drug–target interactions (DTIs) from molecular and protein information can prioritize candidate mechanisms, but apparent accuracy can depend on what information crosses an evaluation split. We developed ArnoldiGCL-S1/S2, a graph-based DTI predictor that combines separate drug and protein projections with seven similarity-neighbour descriptors computed only from positive training-fold interactions. Across BioSNAP, Human and BindingDB under random, cold-drug, cold-protein and cold-pair splits, the full model had the highest mean AUPR among the principal baselines in 7/12 prespecified conditions and exceeded its structural-only variant in 11/12 conditions. The latter gains ranged from 0.013 to 0.396 AUPR; 11 descriptive paired contrasts had Benjamini–Hochberg-adjusted $p < 0.05$. The model was not the highest-AUPR principal baseline in BioSNAP Cold protein, Human Cold protein, Human Cold pair, BindingDB Cold protein, BindingDB Cold pair, demonstrating that the side information does not confer a uniform advantage. These results support leakage-safe similarity-neighbour summaries as a useful, but split-dependent, source of evidence for computational DTI prioritization; they do not establish prospective or experimental target binding.

Index Terms—drug target interaction prediction, graph neural networks, similarity neighbours, data leakage, cold start evaluation

I. Introduction

Inferring drug–target interactions (DTIs) from molecular and protein information can prioritize candidate mechanisms for drug discovery and repurposing. Data-driven predictors have therefore progressed from paired sequence or chemical representations, exemplified by DeepDTA, to graph-based drug encoders and attention-based interaction models [1], [2], [3]. These approaches have established useful representation-learning routes, but their reported performance remains strongly coupled to what evidence is allowed to cross an evaluation split.

This issue is particularly consequential for binary DTI prediction. A model may perform well when a test pair contains a drug or a protein that is well represented in the training graph, while its utility is less clear when a drug, protein or pair is held out. Similarity between compounds or proteins can

be valuable for these settings: earlier predictors have used neighbourhood regularization and heterogeneous network integration to incorporate related drug and protein evidence [4], [5]. However, label-derived neighbour summaries must be constructed from the training partition alone. Otherwise, a held-out DTI label can enter either message passing or a candidate-pair feature indirectly.

Here, we study a leakage-safe use of similarity-neighbour information in a graph-based DTI predictor. The full model, ArnoldiGCL-S1/S2, combines separate drug and protein feature projections, a training-fold structural encoder and seven side-information descriptors derived solely from positive training DTIs. Rather than using similarity only as a regularizer or a network integration mechanism, this design exposes training-fold neighbour evidence as candidate-pair descriptors that can be ablated and shuffled. We evaluate it against feature-only, graph-based and bilinear-attention baselines under fixed random, cold-drug, cold-protein and cold-pair splits. The evaluation is designed to identify not only gains from aligned neighbour information, but also settings in which the same information does not improve over simpler alternatives.

II. Methods

A. Problem setting and feature representations

We considered binary drug–target interaction prediction for a drug set $\mathcal{D}$ and a protein set $\mathcal{P}$. Let $A \in \{0, 1\}^{|\mathcal{D}| \times |\mathcal{P}|}$ denote the observed interaction matrix, where $A_{ij} = 1$ indicates an annotated interaction between drug i and protein j . Each drug was represented by a cached 1,024-bit radius-2 Morgan bit vector [6], and each protein by a cached mean-pooled ESM-2 t30/150M embedding (640 dimensions) [7]. The drug and protein feature encoders were distinct learnable projection networks; their parameters were not shared.

We constructed a drug similarity matrix S_D and a protein similarity matrix S_P by cosine similarity of the

corresponding cached feature vectors. These similarities were feature-derived and were not calculated from DTI labels.

B. Leakage-safe graph construction

For every outer fold, the graph contained only positive DTI edges from the training partition. Drug–drug and protein–protein similarity edges were also restricted to entities observed in that training partition. The default graph connected the top 10 drug neighbours and top 10 protein neighbours; all graph edges were represented bidirectionally. The Threshold GCN baseline instead retained within-type similarity edges whose similarity exceeded 0.3. No validation or test DTI labels were added to a graph or side-information feature.

C. ArnoldiGCL structural representation

The projected drug and protein features were concatenated as nodes in the training-fold graph and passed to the ArnoldiGCL encoder [8]. The structural embedding dimension was 256. The downstream edge scorer received the structural embeddings of a candidate drug–protein pair together with the respective raw Morgan and ESM-2 representations. The encoder was optimized for 50 epochs, and each edge scorer was optimized for 200 epochs with Adam (learning rate 10^{-3} and dropout 0.5). For encoder pretraining, the objective was binary discrimination between the graph representation of the projected node features and a negative view formed by a random permutation of node features. Each update used two positive and two negative node-level logit sets, optimized with binary cross-entropy. The drug and protein projections remained separate throughout this objective.

D. Leakage-safe S1/S2 side information

For a candidate pair (i, j) , all side information was computed from the binary matrix A_{train} that contains positive training DTI edges only. Let $N_D^k(i)$ and $N_P^k(j)$ be the top- k most similar training drugs and proteins, respectively. A queried entity was excluded from its own neighbourhood.

The S1 module used $k = 10$ and produced four descriptors: the mean and standard deviation of $A_{d,j}$ over $d \in N_D^{10}(i)$, and the mean and standard deviation of $A_{d,p}$ over $d \in N_D^{10}(i), p \in N_P^{10}(j)$. Thus, S1 summarizes local neighbour support and its dispersion without accessing the label of the candidate pair.

The S2 module used $k = 20$ and produced three consistency descriptors: the mean of $A_{d,j}$ over $d \in N_D^{20}(i)$, the mean of $A_{i,p}$ over $p \in N_P^{20}(j)$, and the mean of $A_{d,p}$ over the corresponding drug and protein neighbour cross-product. The seven S1/S2 values were passed through the side-information branch of the edge scorer.

The S1-only and S2-only ablations zero-padded the complementary descriptors; the shuffled control permuted the seven-dimensional side-information vectors within each partition while preserving their marginal distribution.

E. Comparators and implementation details

The benchmark includes logistic regression and an MLP on concatenated Morgan and ESM-2 representations, a two-layer GCN on the leak-safe k NN graph, a Threshold GCN comparator, and DrugBAN [1], [2], [3]. All compared methods use the same frozen positive partitions and partition-specific negative examples. This aligns the evaluated drug–protein pairs across methods but does not force identical input modalities: DrugBAN consumes canonical SMILES-derived graphs and protein sequences, whereas the feature and graph comparators use the frozen Morgan and ESM-2 representations described above. DrugBAN is consequently reported as a matched-partition comparator in this benchmark, not as a claim to reproduce its published benchmark scores. For DrugBAN, deterministic SMILES-to-graph and protein integer preprocessing was cached within a fold when an input recurred; this does not alter the DrugBAN architecture, optimizer or emitted batch values. DrugBAN was trained with Adam at a learning rate of 5×10^{-5} , batch size 64 and a fixed 100-epoch schedule; the validation-AUC checkpoint was evaluated once on the corresponding test partition. All DrugBAN folds used TensorFlow-32 matrix operations on NVIDIA GeForce RTX 4080 SUPER GPUs; this execution setting was recorded in every result artifact and held fixed across B5 folds. The two GCN baselines use cached sparse normalized aggregation on BindingDB to bound peak memory. This implementation was verified against the original dense aggregation for forward output and backward gradients, and was smoke-tested on the full BindingDB threshold graph before running the queue.

III. Experimental protocol

A. Datasets and frozen evaluation splits

We evaluate BioSNAP, Human and BindingDB under four scenarios: random, cold-drug, cold-protein and cold-pair splits. Each scenario uses three fixed random seeds and five outer folds, yielding 15 held-out evaluations per dataset–scenario–method combination. Within each outer training partition, approximately 10% of eligible training positives are reserved for model selection. For cold-drug and cold-protein splits, validation and test entities are disjoint from training entities. For cold-pair splits, train, validation and test positives are sampled from disjoint diagonal drug–protein blocks; off-block pairs are not evaluated.

Code-faithful ArnoldiGCL-S1/S2 architecture for DTI prediction

Separate feature projections feed a fold-local graph encoder; pairwise structural, raw-feature and S1/S2 signals are scored jointly.

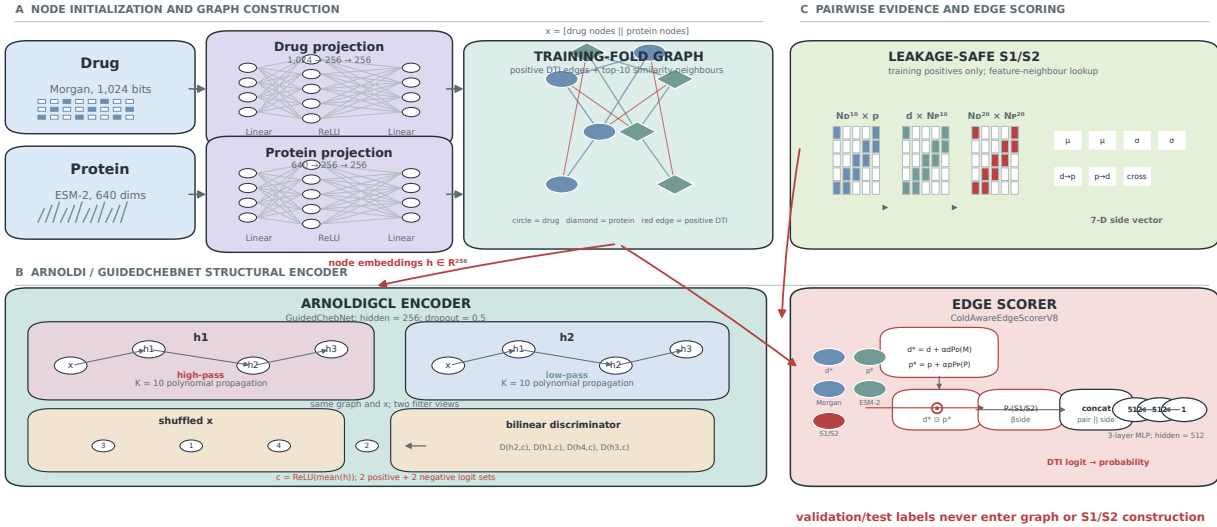


Fig. 1. Overview of the full ArnoldiGCL-S1/S2 DTI predictor. Drug Morgan fingerprints and protein ESM-2 embeddings enter distinct learnable projection networks; the encoder weights are not shared. A training-fold graph supplies structural embeddings, while the S1/S2 branch derives seven candidate-pair descriptors from feature-based neighbours and positive A_{train} edges only. Structural pair features, raw features and side information are then combined by an edge scorer. Validation and test DTI labels are excluded from graph and side-information construction.

The three local full.csv files were verified as exact Git-blob matches to the pinned DrugBAN dataset snapshot (commit 9923f8c99959e00263103ff9ac61ba0eaccc8e02). Its dataset documentation identifies BindingDB, BioSNAP as distributed by MolTrans, and Human as distributed by TransformerCPI [3], [9], [10], [11]. The generated dataset manifest records this source snapshot, byte-level checksums and the observed input schema for every run.

For every partition, up to 10 unique unannotated pairs were sampled per positive without replacement. Test and validation negatives were allocated before training negatives to make held-out samples invariant to later training-set changes. In unusually dense blocks with fewer available unique negatives, all available unique negatives were used and the realized partition size was retained in the fold-level result record.

B. Realized benchmark coverage

The completed release contains 180 frozen split definitions and 1,800 formal result records. This corresponds to 3 datasets, 4 split scenarios, 10 model variants and 15 seed-fold evaluations for every dataset-scenario-method combination. All expected records passed the release audit, and no dataset-scenario-method group had missing folds. The realized test-

set sizes were $30,426 \pm 0$ pairs for BioSNAP Random, $5,793 \pm 6$ pairs for Human Random and $45,483 \pm 5$ pairs for BindingDB Random. Under cold-pair evaluation, the corresponding test-set sizes were $6,173 \pm 864$, $1,168 \pm 411$ and $9,107 \pm 1,732$ pairs, respectively. The larger variation in the cold-start settings reflects the number of eligible entities and unannotated drug-protein pairs available within each frozen partition.

Across the complete benchmark, the mean number of test positives per fold was $2,766 \pm 0$ for BioSNAP Random, 527 ± 1 for Human Random and $4,135 \pm 0$ for BindingDB Random. In the cold-pair setting, the corresponding values were 561 ± 79 , 106 ± 37 and 828 ± 157 . These realized counts, together with the train and validation counts for every dataset and split, are provided in Table II and in the accompanying fold-level source-data CSV.

C. Comparator families

Table I distinguishes feature-only, graph-based, bilinear-attention and ablation controls. All rows use exactly the same frozen positive partitions and partition-specific negative examples. “Threshold GCN” is a comparator implemented with the specified 0.3 within-type similarity threshold; it is not presented as a full reproduction of an external method. The panel is intended for controlled matched-partition

comparisons rather than an exhaustive reproduction of every published DTI method; conclusions are therefore restricted to this benchmark.

D. Evaluation and statistical reporting

We report area under the receiver operating characteristic curve (AUC), area under the precision–recall curve (AUPR), F1 score and accuracy. Checkpoints were selected by validation AUC; F1 thresholds were selected by maximal validation F1 and applied once to the test partition. The primary tables will report mean $\pm$ sample standard deviation across 15 fixed seed–fold evaluations. Pairwise comparisons will be paired by identical seed–fold identity and summarized with effect sizes and two-sided Wilcoxon signed-rank statistics. Because folds sharing a random seed are not independent biological or experimental replicates, these p -values are descriptive summaries of paired fold differences rather than confirmatory population-level inference. For the complete family of reported descriptive comparisons, two-sided p -values will be adjusted using the Benjamini–Hochberg procedure; both raw and adjusted values will be provided as source data.

E. Ablation and robustness analyses

We compare the full S1/S2 model with a structural-only version, S1-only and S2-only variants, and a partition-wise shuffled S1/S2 control. The shuffled control distinguishes the presence of extra numerical inputs from alignment between side information and the evaluated drug–protein pair.

F. Fold-level dataset sizes

Table II reports realized sample counts across the frozen folds. Counts may vary across cold-start folds because entities and eligible unannotated pairs are partition-specific.

G. Compute environment and reproducibility

The principal benchmark was executed on NVIDIA GeForce RTX 4080 SUPER GPUs. Every result artifact records the dataset, scenario, seed, outer fold, split signature, validation criterion, test metrics and realized partition sizes. DrugBAN artifacts additionally record the fixed 100-epoch schedule, input preprocessing cache and TensorFloat-32 setting. The release-time environment manifest records the software versions, GPU/driver information and hashes of the key benchmark scripts. Aggregate wall-clock time is not reported because per-fold duration is not a standardized field in every result artifact.

IV. Results

A. Complete benchmark comparison

Tables III and IV report test performance from the complete frozen-fold benchmark. The accompanying source-data CSV records every seed–fold value and the paired-comparison file contains raw and Benjamini–Hochberg-adjusted Wilcoxon p -values.

Across the 12 prespecified dataset–split conditions, ArnoldiGCL-S1/S2 had the highest mean AUPR among the principal baselines in 7/12 conditions. The conditions without the highest mean AUPR were BioSNAP Cold protein, Human Cold protein, Human Cold pair, BindingDB Cold protein, BindingDB Cold pair.

B. Contribution of S1 and S2

Table V reports the prespecified structural-only, single-module and shuffled controls. Relative to the structural-only variant, the full S1/S2 model had higher mean AUPR in 11/12 conditions; among these, 11 had a descriptive paired Wilcoxon contrast with BH-adjusted $p < 0.05$. The positive mean differences ranged from 0.013 to 0.396. The full model was not higher than the structural-only variant in Human Cold pair.

C. Where the method does not improve

The complete source data, rather than a selected subset of splits, are used for all comparative statements. In particular, the principal-baseline comparison did not place the full model first in BioSNAP Cold protein, Human Cold protein, Human Cold pair, BindingDB Cold protein, BindingDB Cold pair. These settings are retained in the figures, tables and Discussion rather than being omitted from the main comparison.

V. Discussion

The fixed-fold benchmark indicates that similarity-neighbour information can add predictive signal beyond the training-fold structural encoder, but only when its construction is constrained to the training partition. The full S1/S2 model had higher mean AUPR than the structural-only variant in 11/12 conditions, and 11 descriptive paired contrasts had a Benjamini–Hochberg-adjusted $p < 0.05$. This pattern is consistent with the interpretation that local neighbourhood support and drug–protein consistency can complement graph-derived pair representations.

The shuffled control provides a narrower interpretation of this association. The full model had higher mean AUPR than the shuffled S1/S2 control in 12/12 conditions, including 11 descriptive paired contrasts with FDR-adjusted $p < 0.05$. Because shuffling preserves the marginal side-feature values while breaking

Family	Model	Information available to the model
Feature-only	Logistic regression	Concatenated Morgan and ESM-2 pair features.
Feature-only	MLP	Concatenated Morgan and ESM-2 pair features.
Graph-based	GCN	Fold-local positive DTI graph with top-10 within-type similarity neighbours.
Graph-based	Threshold GCN	Fold-local positive DTI graph with within-type similarity edges above 0.3.
Bilinear-attention	DrugBAN	Canonical molecular graph and protein-sequence inputs, trained on the same frozen pair partitions.
Ablation	ArnoldiGCL w/o S1/S2; +S1; +S2; shuffled S1/S2	The same fold-local ArnoldiGCL structural encoder, with the indicated side-information treatment.
Full model	ArnoldiGCL-S1/S2	Structural pair features, raw Morgan and ESM-2 features, and aligned seven-dimensional S1/S2 information.

TABLE I

Prespecified comparator families. All learning, graph construction and label-derived side information are restricted to the outer training fold.

Dataset	Split	Train pairs	Validation pairs	Test pairs	Test positives
BioSNAP	Random	109538 $\pm$ 0	12166 $\pm$ 0	30426 $\pm$ 0	2766 $\pm$ 0
BioSNAP	Cold drug	109724 $\pm$ 1569	11980 $\pm$ 750	30426 $\pm$ 1336	2766 $\pm$ 121
BioSNAP	Cold protein	109120 $\pm$ 4026	12584 $\pm$ 3126	30426 $\pm$ 3839	2766 $\pm$ 349
BioSNAP	Cold pair	78327 $\pm$ 3469	1021 $\pm$ 315	6173 $\pm$ 864	561 $\pm$ 79
Human	Random	20849 $\pm$ 6	2321 $\pm$ 0	5793 $\pm$ 6	527 $\pm$ 1
Human	Cold drug	21054 $\pm$ 989	2116 $\pm$ 912	5793 $\pm$ 1194	527 $\pm$ 109
Human	Cold protein	20789 $\pm$ 1099	2382 $\pm$ 843	5793 $\pm$ 1139	527 $\pm$ 104
Human	Cold pair	15101 $\pm$ 1453	150 $\pm$ 112	1168 $\pm$ 411	106 $\pm$ 37
BindingDB	Random	163737 $\pm$ 5	18194 $\pm$ 0	45483 $\pm$ 5	4135 $\pm$ 0
BindingDB	Cold drug	163701 $\pm$ 700	18230 $\pm$ 466	45483 $\pm$ 685	4135 $\pm$ 62
BindingDB	Cold protein	163327 $\pm$ 8143	18605 $\pm$ 4576	45483 $\pm$ 8652	4135 $\pm$ 787
BindingDB	Cold pair	118036 $\pm$ 8799	1481 $\pm$ 538	9107 $\pm$ 1732	828 $\pm$ 157

TABLE II

Realized fold sizes (mean $\pm$ sample standard deviation across 15 fixed seed-fold evaluations).

Dataset	Split	Logistic regression	MLP	GCN	Threshold GCN	DrugBAN	M8 + S1+S2
BioSNAP	Random	0.458 $\pm$ 0.012	0.746 $\pm$ 0.008	0.618 $\pm$ 0.039	0.633 $\pm$ 0.016	0.759 $\pm$ 0.010	0.831 $\pm$ 0.007
BioSNAP	Cold drug	0.376 $\pm$ 0.019	0.583 $\pm$ 0.017	0.529 $\pm$ 0.022	0.551 $\pm$ 0.025	0.675 $\pm$ 0.017	0.722 $\pm$ 0.015
BioSNAP	Cold protein	0.303 $\pm$ 0.044	0.634 $\pm$ 0.030	0.212 $\pm$ 0.031	0.250 $\pm$ 0.070	0.322 $\pm$ 0.052	0.597 $\pm$ 0.034
BioSNAP	Cold pair	0.177 $\pm$ 0.034	0.367 $\pm$ 0.034	0.166 $\pm$ 0.021	0.203 $\pm$ 0.037	0.197 $\pm$ 0.039	0.394 $\pm$ 0.036
Human	Random	0.557 $\pm$ 0.027	0.685 $\pm$ 0.019	0.765 $\pm$ 0.018	0.584 $\pm$ 0.054	0.766 $\pm$ 0.021	0.768 $\pm$ 0.018
Human	Cold drug	0.166 $\pm$ 0.022	0.529 $\pm$ 0.059	0.541 $\pm$ 0.052	0.498 $\pm$ 0.109	0.579 $\pm$ 0.091	0.588 $\pm$ 0.075
Human	Cold protein	0.257 $\pm$ 0.050	0.500 $\pm$ 0.123	0.503 $\pm$ 0.055	0.506 $\pm$ 0.070	0.458 $\pm$ 0.067	0.496 $\pm$ 0.083
Human	Cold pair	0.122 $\pm$ 0.039	0.195 $\pm$ 0.052	0.149 $\pm$ 0.107	0.160 $\pm$ 0.062	0.177 $\pm$ 0.084	0.178 $\pm$ 0.065
BindingDB	Random	0.355 $\pm$ 0.004	0.897 $\pm$ 0.003	0.615 $\pm$ 0.010	0.667 $\pm$ 0.010	0.925 $\pm$ 0.004	0.947 $\pm$ 0.003
BindingDB	Cold drug	0.352 $\pm$ 0.011	0.891 $\pm$ 0.010	0.614 $\pm$ 0.024	0.673 $\pm$ 0.014	0.909 $\pm$ 0.010	0.928 $\pm$ 0.007
BindingDB	Cold protein	0.099 $\pm$ 0.009	0.494 $\pm$ 0.036	0.108 $\pm$ 0.010	0.111 $\pm$ 0.016	0.250 $\pm$ 0.053	0.364 $\pm$ 0.100
BindingDB	Cold pair	0.093 $\pm$ 0.006	0.467 $\pm$ 0.054	0.095 $\pm$ 0.015	0.098 $\pm$ 0.013	0.234 $\pm$ 0.058	0.373 $\pm$ 0.077

TABLE III

Test AUPR for the principal baseline comparison. Values are mean $\pm$ sample standard deviation across 15 fixed seed-fold evaluations.

their alignment with each candidate pair, the comparison suggests that aligned side information, rather than only feature scale or dimensionality, contributes to the observed differences. It does not by itself establish a biological mechanism for any predicted interaction.

The result is not a claim of uniform superiority. The full model was not the highest-AUPR principal baseline in BioSNAP Cold protein, Human Cold protein, Human Cold pair, BindingDB Cold protein, BindingDB Cold pair. These settings should be treated as part of the method boundary: a similarity-neighbour summary may be less informative when representation coverage, dataset composition or the held-out entity structure differs from the conditions in which it is

advantageous.

Several limitations follow directly from the evaluation design. The labels are curated DTIs, and the capped samples of unannotated pairs may include unknown positives; the reported metrics therefore describe the specified benchmark sampling distributions rather than the prevalence of interactions in a complete chemical-proteomic space. The study uses fixed molecular and protein feature caches, rather than learning representations from raw structures or sequences within each fold. Finally, the benchmark evaluates ranking and classification against annotated labels; experimental binding assays and prospective candidate selection are required before a prediction

Baseline comparison

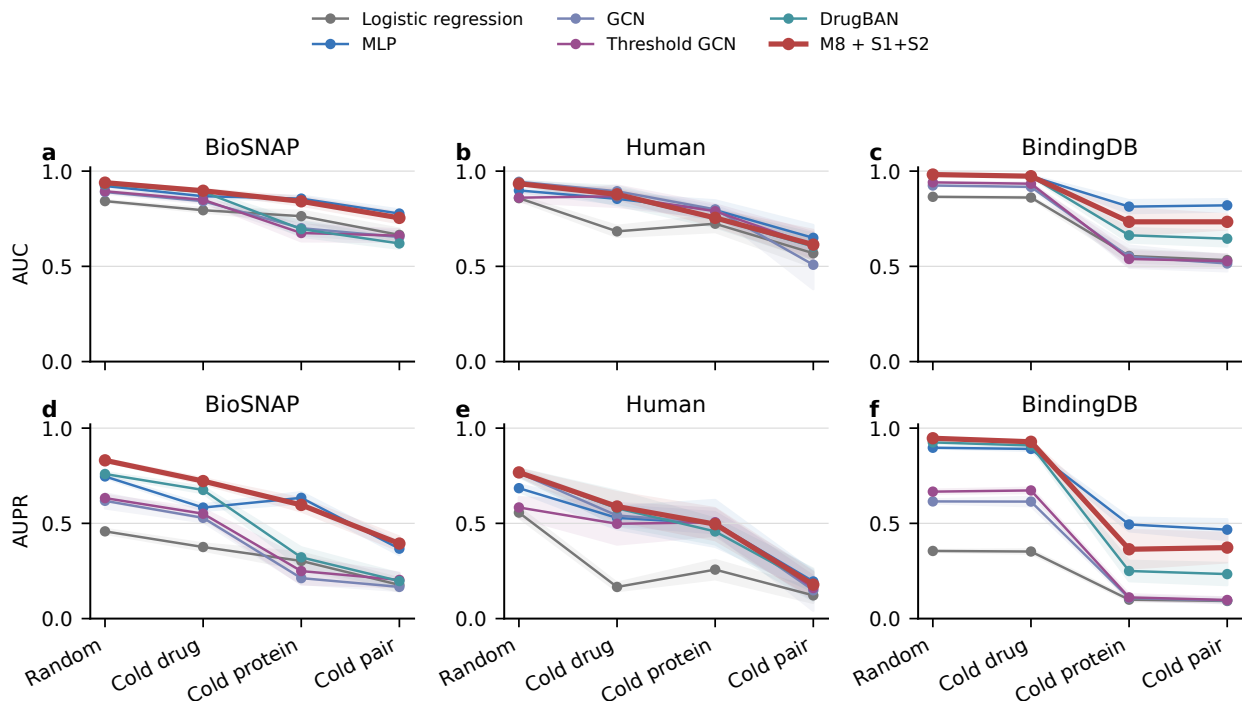


Fig. 2. Comparison of feature-only, graph-based, bilinear-attention and full S1/S2 models across the four fixed split scenarios. Each point is the mean of 15 seed-fold evaluations and the shaded band denotes the sample standard deviation.

Dataset	Split	Logistic regression	MLP	GCN	Threshold GCN	DrugBAN	M8 + S1+S2
BioSNAP	Random	0.842 $\pm$ 0.005	0.922 $\pm$ 0.005	0.890 $\pm$ 0.008	0.894 $\pm$ 0.006	0.930 $\pm$ 0.003	0.939 $\pm$ 0.004
BioSNAP	Cold drug	0.795 $\pm$ 0.011	0.868 $\pm$ 0.008	0.842 $\pm$ 0.013	0.850 $\pm$ 0.013	0.892 $\pm$ 0.008	0.897 $\pm$ 0.008
BioSNAP	Cold protein	0.763 $\pm$ 0.028	0.856 $\pm$ 0.015	0.700 $\pm$ 0.036	0.675 $\pm$ 0.043	0.696 $\pm$ 0.040	0.841 $\pm$ 0.022
BioSNAP	Cold pair	0.665 $\pm$ 0.032	0.777 $\pm$ 0.021	0.654 $\pm$ 0.027	0.661 $\pm$ 0.032	0.620 $\pm$ 0.027	0.755 $\pm$ 0.028
Human	Random	0.858 $\pm$ 0.012	0.899 $\pm$ 0.009	0.944 $\pm$ 0.007	0.860 $\pm$ 0.023	0.939 $\pm$ 0.008	0.934 $\pm$ 0.006
Human	Cold drug	0.684 $\pm$ 0.028	0.854 $\pm$ 0.027	0.894 $\pm$ 0.014	0.868 $\pm$ 0.056	0.871 $\pm$ 0.050	0.878 $\pm$ 0.025
Human	Cold protein	0.724 $\pm$ 0.043	0.796 $\pm$ 0.050	0.799 $\pm$ 0.018	0.789 $\pm$ 0.036	0.751 $\pm$ 0.045	0.755 $\pm$ 0.043
Human	Cold pair	0.569 $\pm$ 0.072	0.650 $\pm$ 0.067	0.508 $\pm$ 0.128	0.610 $\pm$ 0.069	0.604 $\pm$ 0.067	0.614 $\pm$ 0.077
BindingDB	Random	0.865 $\pm$ 0.002	0.978 $\pm$ 0.001	0.924 $\pm$ 0.002	0.941 $\pm$ 0.002	0.979 $\pm$ 0.002	0.982 $\pm$ 0.001
BindingDB	Cold drug	0.861 $\pm$ 0.005	0.973 $\pm$ 0.004	0.917 $\pm$ 0.005	0.934 $\pm$ 0.005	0.973 $\pm$ 0.004	0.973 $\pm$ 0.003
BindingDB	Cold protein	0.554 $\pm$ 0.034	0.814 $\pm$ 0.034	0.551 $\pm$ 0.058	0.539 $\pm$ 0.040	0.663 $\pm$ 0.037	0.734 $\pm$ 0.071
BindingDB	Cold pair	0.533 $\pm$ 0.028	0.820 $\pm$ 0.032	0.515 $\pm$ 0.041	0.527 $\pm$ 0.035	0.645 $\pm$ 0.040	0.734 $\pm$ 0.039

TABLE IV

Test AUC for the principal baseline comparison. Values are mean $\pm$ sample standard deviation across 15 fixed seed-fold evaluations.

can support a pharmacological claim.

VI. Conclusion

ArnoldiGCL-S1/S2 integrates separate drug and protein projections, a training-fold graph encoder and leakage-safe S1/S2 similarity-neighbour descriptors for DTI prediction. Under the complete fixed-fold benchmark, the full model exceeded its structural-only counterpart in 11/12 AUPR comparisons and was the highest-AUPR principal baseline in 7/12 conditions. The evidence supports the use of fold-local neigh-

bour information as a split-dependent computational prioritization signal, while preserving the observed non-winning settings and the need for prospective experimental validation.

VII. Data and code availability

Generated benchmark artifacts. The frozen train, validation and test partitions, per-fold result JSON files, aggregate tables, figure source data and the dataset manifest generated in this study will be deposited in the authors' public repository with a persistent

Ablation of S1/S2 side information

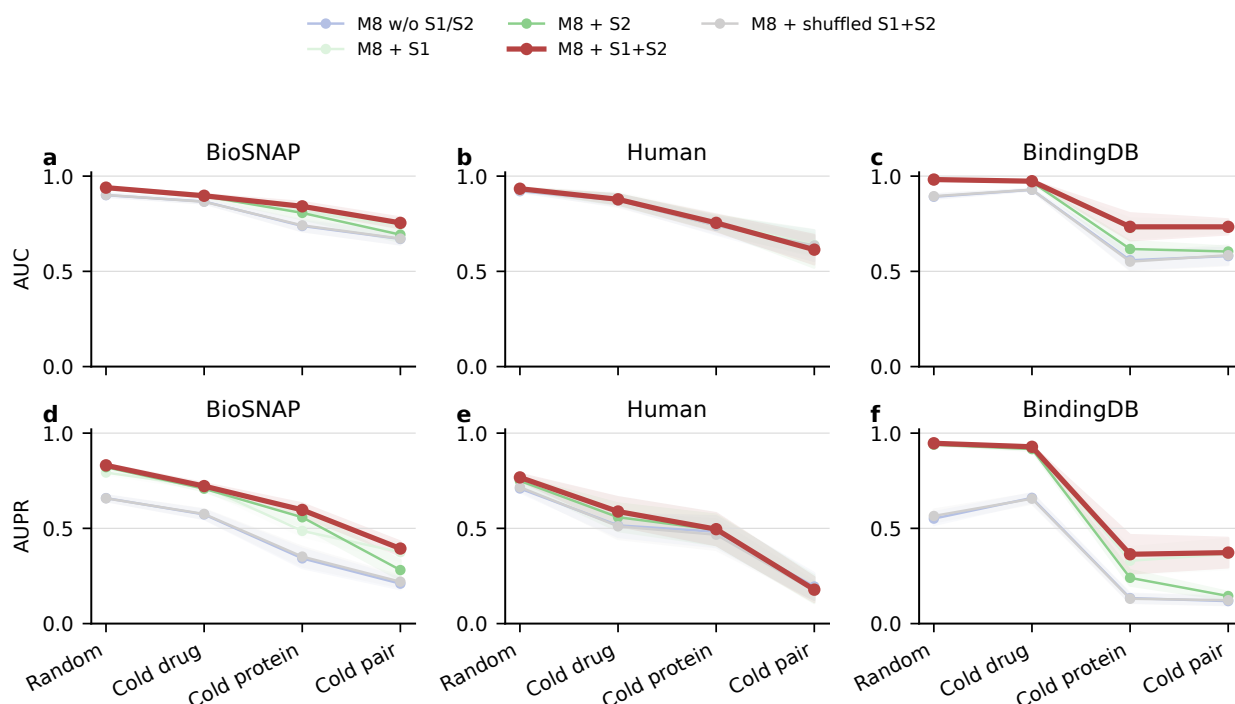


Fig. 3. Ablation of S1/S2 side information. The shuffled control preserves the marginal values of the seven descriptors within each partition but disrupts their association with candidate drug–protein pairs. Each point is the mean of 15 seed–fold evaluations and the shaded band denotes the sample standard deviation.

identifier before publication. The manifest records the SHA256 checksum, Git-blob checksum, observed schema and pinned upstream snapshot of every input CSV used for the benchmark.

Reused input data. The benchmark uses local CSV files labelled BioSNAP, Human and BindingDB. Each file was verified byte-for-byte against the corresponding full.csv file in the DrugBAN repository dataset snapshot (commit 9923f8c99959e00263103ff9ac61ba0eaccc8e02); that snapshot documents BindingDB, MolTrans/BioSNAP and TransformerCPI/Human as the respective sources [3], [9], [10], [11]. The DrugBAN repository is MIT-licensed; original data-provider terms govern reuse and redistribution, so this manuscript does not claim that the local copies may be redistributed.

Code availability. The frozen split generator, benchmark runners, plotting scripts, manuscript generators and final audit script will be made available in the authors’ public code repository at the release tag associated with this manuscript. The release will include environment specifications and commands required to regenerate the reported tables and figures from the archived result artifacts.

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Declarations

Author contributions. Laiyi Fu conducted the study and prepared the manuscript. Jianyu Yin contributed to the methodology and manuscript review.

Competing interests. The authors declare no competing interests.

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Dataset	Split	M8 w/o S1/S2	M8 + S1	M8 + S2	M8 + S1+S2	M8 + shuffled S1+S2
BioSNAP	Random	0.659 ± 0.015	0.793 ± 0.008	0.821 ± 0.008	0.831 ± 0.007	0.658 ± 0.014
BioSNAP	Cold drug	0.573 ± 0.024	0.721 ± 0.017	0.708 ± 0.014	0.722 ± 0.015	0.575 ± 0.024
BioSNAP	Cold protein	0.342 ± 0.050	0.487 ± 0.027	0.558 ± 0.049	0.597 ± 0.034	0.352 ± 0.051
BioSNAP	Cold pair	0.210 ± 0.029	0.375 ± 0.047	0.281 ± 0.052	0.394 ± 0.036	0.220 ± 0.029
Human	Random	0.709 ± 0.019	0.766 ± 0.015	0.749 ± 0.020	0.768 ± 0.018	0.716 ± 0.017
Human	Cold drug	0.516 ± 0.060	0.589 ± 0.072	0.558 ± 0.072	0.588 ± 0.075	0.511 ± 0.066
Human	Cold protein	0.482 ± 0.081	0.492 ± 0.088	0.492 ± 0.079	0.496 ± 0.083	0.467 ± 0.079
Human	Cold pair	0.195 ± 0.070	0.184 ± 0.058	0.180 ± 0.072	0.178 ± 0.065	0.172 ± 0.047
BindingDB	Random	0.551 ± 0.033	0.941 ± 0.004	0.939 ± 0.004	0.947 ± 0.003	0.565 ± 0.033
BindingDB	Cold drug	0.660 ± 0.023	0.928 ± 0.006	0.916 ± 0.008	0.928 ± 0.007	0.656 ± 0.026
BindingDB	Cold protein	0.134 ± 0.023	0.330 ± 0.070	0.240 ± 0.040	0.364 ± 0.100	0.130 ± 0.021
BindingDB	Cold pair	0.118 ± 0.019	0.365 ± 0.078	0.145 ± 0.021	0.373 ± 0.077	0.123 ± 0.026

TABLE V

Test AUPR for the S1/S2 ablations and shuffled control. Values are mean $\pm$ sample standard deviation across 15 fixed seed-fold evaluations.

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